Optimal Treatment Assignment Designs for Spatial Cluster Randomized Trials

Simulation Study — Project Summary

Andrew Walther

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1 Executive Summary

This report summarizes a completed simulation study that systematically compares eight treatment assignment strategies for Spatial Cluster Randomized Trials (CRTs). The study generated 12,800 scenarios by sweeping the direct treatment effect (τ) and spatial DGP parameters across realistic ranges, using Maximum Likelihood Estimation (MLE) to evaluate each design. Key findings:

- **Saturation Quadrants and Incidence-Guided Saturation Quadrants are the clear winners** across all conditions tested. At the primary effect size ($\tau = 1.0$), Incidence-Guided Saturation Quadrants achieves MSE = 0.079 and Saturation Quadrants achieves MSE = 0.080 — approximately **ten times lower** than the worst-performing Block Stratified Sampling (MSE = 0.802).
- **Spatial separation is counterproductive.** Block Stratified Sampling, which maximizes spatial distance between treated and control clusters, performs dramatically worse than all alternatives. Its 95% confidence intervals contain the true effect only ~55% of the time — far below the nominal 95% target.
- **Saturation-based designs are robust.** The saturation design ranking advantage holds across all tested levels of spatial autocorrelation (ρ), spillover magnitude (γ), spillover direction, and neighborhood structure. Two parameters — spillover direction and contiguity type — had no meaningful effect on rankings and are noted only briefly.
- **Coverage and power are strong for recommended designs.** MLE achieves approximately 94% coverage for all designs except Block Stratified Sampling across the full effect size range ($\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$), with Block Stratified Sampling remaining severely anti-conservative throughout.
- **Path forward:** These results motivate applying Saturation Quadrants and Incidence-Guided Saturation Quadrants to real-world NC county-level Sudden Unexpected Death (SUD) data, where spatial dependence in outcomes is well-established and design choice carries substantial practical consequence.

2 Background and Motivation

2.1 What Is a Spatial CRT?

A Cluster Randomized Trial (CRT) assigns treatment at the level of groups — called clusters — rather than individuals. In a **spatial CRT**, clusters are geographically defined units (e.g., counties, census tracts, health districts), and outcomes may exhibit two forms of spatial structure:

1. **Spatial autocorrelation:** Nearby clusters tend to have similar outcomes, independent of treatment, due to shared socioeconomic conditions, demographics, or infrastructure.
2. **Spillover (interference):** Treatment applied to one cluster can affect outcomes in neighboring clusters — for example, when an intervention reduces disease transmission or shifts behavior across geographic boundaries.

Standard CRT analysis assumes clusters are exchangeable and independent. When spatial structure is present, this assumption fails: a poorly chosen design can create systematic bias and grossly miscalibrated confidence intervals, even when estimation is carried out with a sophisticated spatial model.

2.2 Application Context

This project is motivated by evaluating interventions targeting **Sudden Unexpected Death (SUD)** in North Carolina counties. SUD incidence in NC follows a Poisson process with a base rate of approximately 35 events per 100,000 person-years (Mirzaei et al.), and outcomes across neighboring counties are plausibly correlated through shared risk factors and service networks.

Selecting a treatment assignment design that accounts for this spatial structure is critical. As shown below, choosing the wrong design — even when paired with a correct spatial estimator — can inflate the mean squared error of the treatment effect estimate by a factor of ten.

2.3 Why Eight Designs?

Several intuitive strategies exist for assigning treatment in a spatial CRT: concentrate treatment in high-burden areas, separate treated and control clusters as widely as possible, randomize within geographic blocks, or vary treatment saturation across spatial regions. This study systematically compares eight such strategies under a common, realistic data-generating process, providing an empirical basis for principled design selection.

3 Methods

3.1 Study Region and Spatial Structure

The simulated study region is a **10×10 regular lattice** of 100 clusters. Spatial adjacency is defined by either rook contiguity (shared edges; 4 neighbors per interior cluster) or queen contiguity (shared edges or corners; 8 neighbors per interior cluster), yielding a row-standardized spatial weight matrix $\mathbf{W}$. Results are robust to the choice of contiguity definition — this parameter had no meaningful effect on design rankings — and is not discussed further.

3.2 Outcome Data Generating Process

Outcomes are generated by a **Spatial Durbin Model (SDM)**:

$$\mathbf{Y} = \tau\mathbf{Z} + \gamma\mathbf{WZ} + \beta\mathbf{X} + \rho\mathbf{WY} + \varepsilon$$

Each term is defined in Table 1. The term $\gamma\mathbf{WZ}$ captures **average treatment intensity among neighbors** — the mechanism by which treatment spills over into adjacent clusters. The term $\rho\mathbf{WY}$ induces spatial autocorrelation in the outcome itself, independent of treatment.

Baseline incidence ($\mathbf{X}$) is generated under two modes, representing realistic outcome distributions:

- *Spatial:* Uniform marginal values with spatial autocorrelation ($\rho_X \in \{0.20, 0.50\}$)
- *Poisson:* Count-based SUD-realistic incidence ($\lambda = 35/100{,}000$, $\rho_X \in \{0.20, 0.50\}$)

Table 1: SDM parameter definitions.

Symbol	Definition
Y	$N \times 1$ vector of cluster outcomes
τ	Direct treatment effect (estimand of interest)
Z	$N \times 1$ binary treatment assignment vector
γ	Spillover coefficient — effect of neighbors' treatment on own outcome
W	$N \times N$ row-standardized spatial weight matrix
β	Incidence loading coefficient (fixed at 1.0)
X	$N \times 1$ baseline incidence vector
ρ	Outcome spatial autocorrelation
ε	$N \times 1$ i.i.d. residuals, $\varepsilon_i \sim N(0, \sigma^2)$

The full parameter sweep covers **12,800 scenarios** across four incidence configurations, as summarized in Table 2.

Table 2: Simulation parameter grid. Each combination of the five swept parameters was fully crossed, yielding 12,800 total scenarios.

Parameter	Values	Levels
Direct treatment effect (τ)	0.8, 1.0, 1.5, 2.0, 3.0	5
Outcome spatial autocorrelation (ρ)	0.00, 0.01, 0.20, 0.50	4
Spillover magnitude (γ)	0.5, 0.6, 0.7, 0.8	4
Spillover direction	Control only; both directions	2
Incidence config (mode $\times$ ρ_X)	Spatial $\times$ 0.20, 0.50; Poisson $\times$ 0.20, 0.50	4

3.3 The Eight Treatment Assignment Designs

Figure 1 shows all eight designs applied to the same 10×10 grid instance, with cluster color indicating baseline incidence and within-quadrant saturation percentages annotated for Saturation Quadrants and Incidence-Guided Saturation Quadrants.

Table 3: The eight treatment assignment strategies evaluated, grouped by conceptual design type. The ID column shows simulation indices; display order reflects conceptual grouping (Blocking → Stratified → Saturation) rather than ID order. All designs target approximately 50% treated clusters. Deterministic designs produce the same assignment for every replicate given fixed baseline incidence.

ID	Name	Description	Det.
1	Block Stratified Sampling	Alternating grid cells — maximizes spatial separation between treated and control	Yes
5	2x2 Blocking	1:1 randomization within non-overlapping 2x2 spatial blocks	No
4	Isolation Buffer	Greedy assignment: no two treated clusters share a border	No
2	High Incidence Focus	Treats all top-50% burden clusters by baseline incidence	Yes
7	Balanced Halves	2-strata median split; balanced 1:1 assignment within each half	No
6	Balanced Quartiles	Stratified randomization within incidence quartile strata	No
3	Saturation Quadrants	Randomizes saturation rate (20-80%) independently within each geographic quadrant	No
8	Incidence-Guided Saturation Quadrants	Quadrant saturation rates guided by average incidence rank per quadrant	No

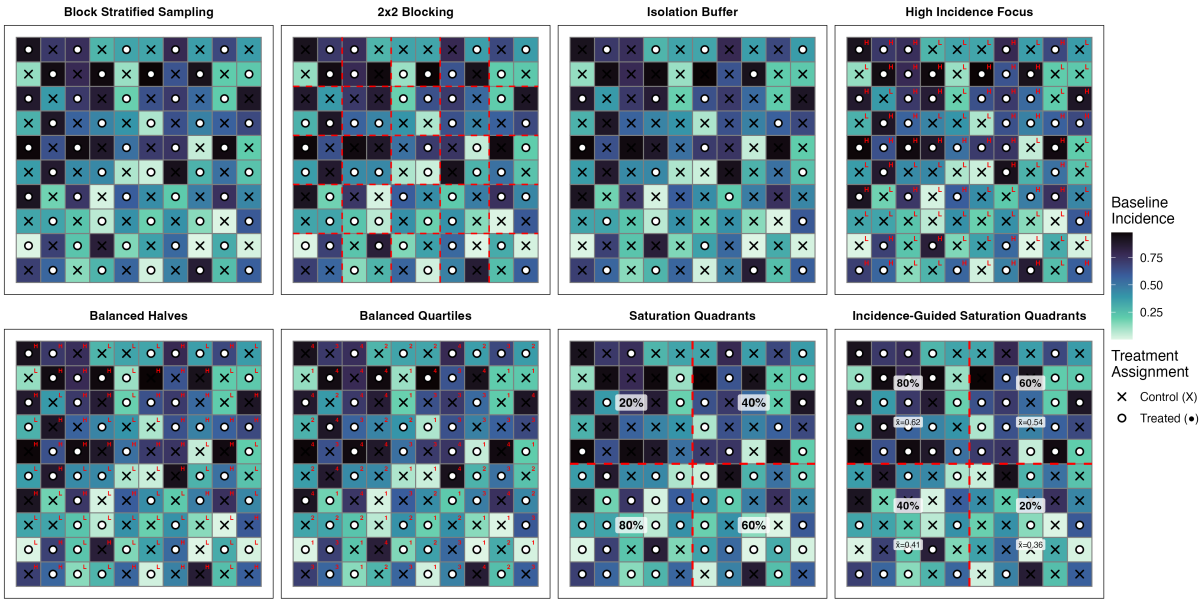

Figure 1: All eight treatment assignment designs on a common 10x10 grid, grouped by conceptual type. Cell shading indicates baseline incidence; filled circles are treated clusters, X marks are control. Saturation percentages indicate the fraction of clusters treated within each geographic quadrant for Saturation Quadrants and Incidence-Guided Saturation Quadrants; mean quadrant incidence (x-bar) is also shown for the latter.

The conceptual contrast across designs is important: Block Stratified Sampling enforces **maximum spatial separation** between treated and control clusters. Saturation Quadrants and Incidence-Guided Saturation Quadrants take the opposite approach — they vary **saturation intensity** within spatial regions, intentionally mixing treated and control units at a local level. This distinction directly determines how well each design accommodates spatial spillover in the outcome DGP.

3.4 Estimation and Evaluation Metrics

All results use **Maximum Likelihood Estimation (MLE)** via the `lagsarlm` function from the `spatialreg` package. The model is fit as an oracle — the true spillover term (**WZ**) is included as a covariate, so MLE has access to the correct spillover structure. Each scenario uses 25 design resamples $\times$ 10 outcome resamples = **250 Monte Carlo iterations**.

Four metrics are tracked per scenario:

$$\text{Bias} = \mathbb{E}[\hat{\tau}] - \tau \qquad \text{MSE} = \text{Bias}^2 + \text{Var}(\hat{\tau}) \qquad \text{Coverage} = \mathbb{P}\left(\tau \in \widehat{\text{CI}}_{95\%}\right) \qquad \text{Power} = \mathbb{P}(\text{reject } H_0 : \tau = 0)$$

MSE is the primary criterion for design comparison. Coverage provides an inference diagnostic — a well-designed trial should achieve approximately 95% coverage regardless of the true effect size.

4 Results

4.1 Overall Design Performance

Table 4 summarizes design performance at the primary effect size ($\tau = 1.0$), averaged across all four Spatial and Poisson incidence configurations.

Table 4: Mean MSE and coverage at the primary effect size (tau = 1.0), averaged across Spatial and Poisson incidence configurations. Designs are sorted by MSE (best to worst).

Design	Mean MSE	Mean Coverage	Mean Power
Incidence Sat. Quad.	**0.079**	**0.940**	**0.950**
Sat. Quadrants	**0.080**	**0.938**	**0.945**
Bal. Quartiles	0.103	0.939	0.881
Bal. Halves	0.110	0.938	0.869
Hi-Inc. Focus	0.146	0.964	0.886
Iso. Buffer	0.148	0.939	0.776
2x2 Blocking	0.228	0.938	0.729
Block Stratified	0.738	0.541	0.477

> **Note**
>
> **Note for manuscript development:** Saturation Quadrants and Incidence-Guided Saturation Quadrants perform similarly (MSE 0.079 vs. 0.080 at = 1.0), as do Balanced Quartiles and Balanced Halves. For final manuscript reporting, one member of each pair will be selected

as the representative design to streamline presentation. This decision will be deferred until the application to NC county data is complete.

Saturation Quadrants and Incidence-Guided Saturation Quadrants are the top performers by a substantial margin; Block Stratified Sampling is dramatically inferior. To understand *why*, Figure 2 decomposes each design's MSE into its squared bias and variance components for the most SUD-realistic configuration (Poisson incidence, $\rho_X = 0.20$).

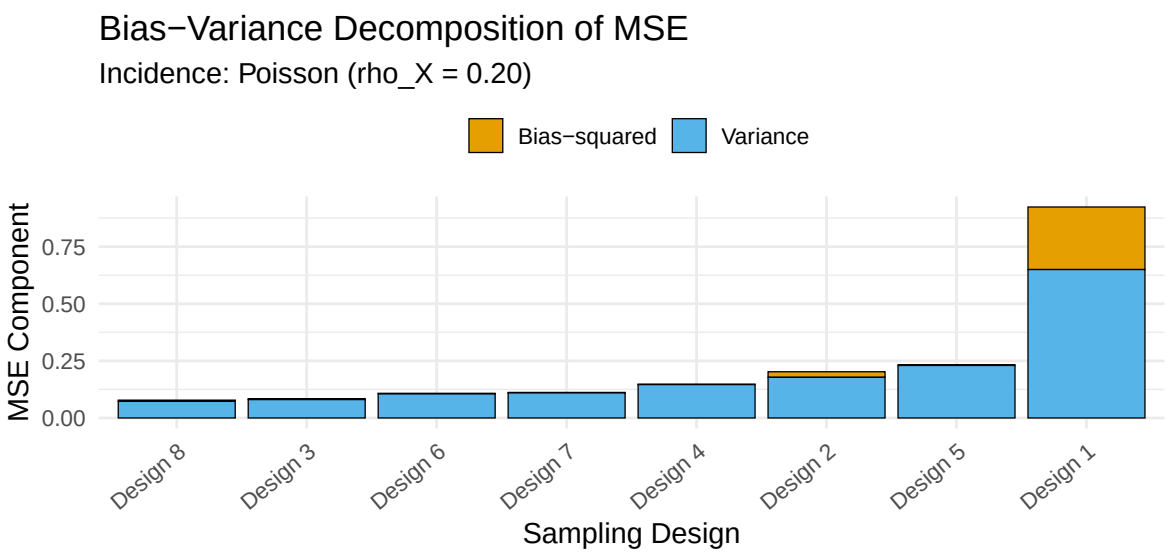

Figure 2: Bias-squared and variance contributions to MSE for each design (Poisson incidence, rho_X = 0.20, tau = 1.0). Block Stratified Sampling's MSE is almost entirely variance; Saturation Quadrants and Incidence-Guided Saturation Quadrants achieve low bias and low variance simultaneously.

Block Stratified Sampling fails because its rigid alternating structure creates near-perfect collinearity between treatment assignment ($\mathbf{Z}$) and average neighbor treatment ($\mathbf{WZ}$). MLE cannot disentangle the direct treatment effect from spillover, producing high-variance estimates. Isolation Buffer suffers a similar problem — its greedy no-adjacency constraint produces sparse, fragmented treatment that is similarly misaligned with the spatial DGP.

Saturation Quadrants and Incidence-Guided Saturation Quadrants succeed precisely because their saturation-based approach distributes treated and control clusters heterogeneously within each quadrant, preserving contrast between $\mathbf{Z}$ and $\mathbf{WZ}$ and keeping both bias and variance low.

4.2 Formal Statistical Comparisons

A **Friedman test** — the non-parametric analogue of a repeated-measures ANOVA, with each simulation scenario treated as a block — confirms that design differences are highly significant: $\chi^2(7) = 1091.90$, $p < 2.2 \times 10^{-16}$. This finding is not specific to any particular parameter setting: stratifying the analysis by ρ, γ, spillover direction, neighbor type, or incidence mode produces

statistically significant design differences in every stratum (all conditional Friedman $p < 2.2 \times 10^{-16}$).

Post-hoc Nemenyi pairwise comparisons (Figure 3, Figure 4) reveal a **four-tier structure** among designs. Saturation Quadrants and Incidence-Guided Saturation Quadrants form a **statistically distinct top tier** — both are significantly different from all other designs ($p < 0.05$). Below them, Balanced Quartiles, High Incidence Focus, and Balanced Halves form a statistically equivalent second tier: the three designs are not significantly different from one another but are significantly worse than the saturation designs. Isolation Buffer and 2×2 Blocking form a third tier, equivalent to each other but significantly worse than the second tier. Block Stratified Sampling stands alone at the bottom, significantly different from every other design.

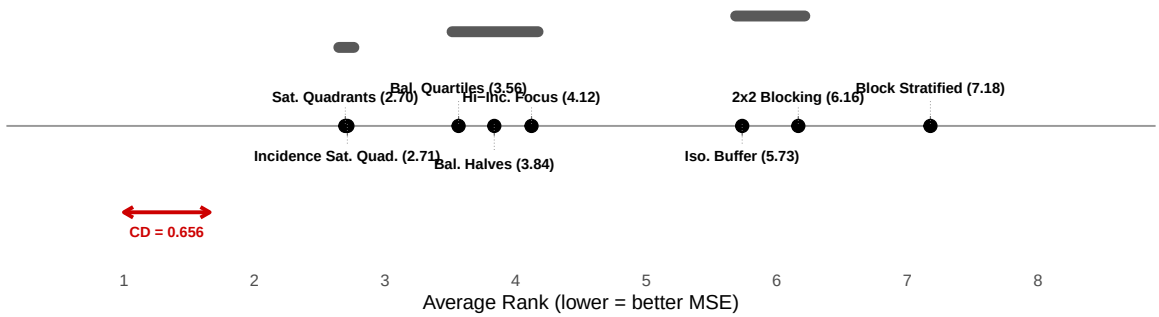

Figure 3: Critical difference (CD) diagram by average rank ($\tau = 1.0$, all incidence configurations). Designs are positioned by their average Friedman rank (lower = better). The red arrow marks the critical difference ($CD = 0.656$): designs whose average ranks differ by less than this threshold are connected by a bold segment and are not significantly different (Nemenyi, $\alpha = 0.05$). Four distinct tiers are visible: Saturation Quadrants and Incidence-Guided Saturation Quadrants (top); Balanced Quartiles, High Incidence Focus, and Balanced Halves (second); Isolation Buffer and 2×2 Blocking (third); Block Stratified Sampling (bottom, isolated).

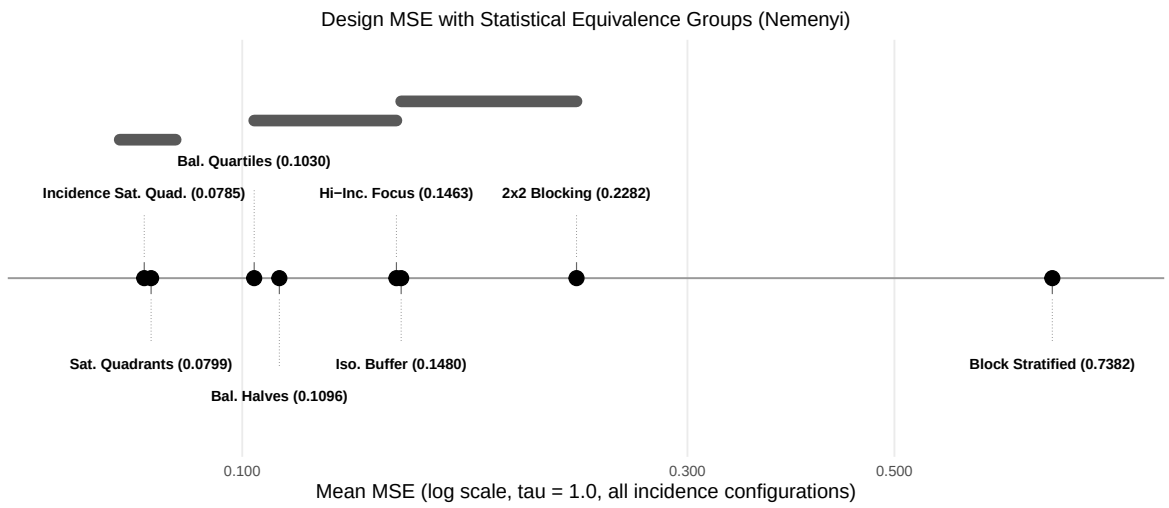

Figure 4: CD diagram by mean MSE ($\tau = 1.0$, all incidence configurations). Each point shows a design's mean MSE on a log scale; designs are sorted left-to-right by increasing MSE. Horizontal bars connect designs that are not significantly different from each other (Nemenyi, $\alpha = 0.05$). This view conveys the practical magnitude of MSE differences alongside the statistical grouping. The four-tier structure from Figure 3 is preserved: the saturation designs achieve the lowest MSE and are clearly separated from all others.

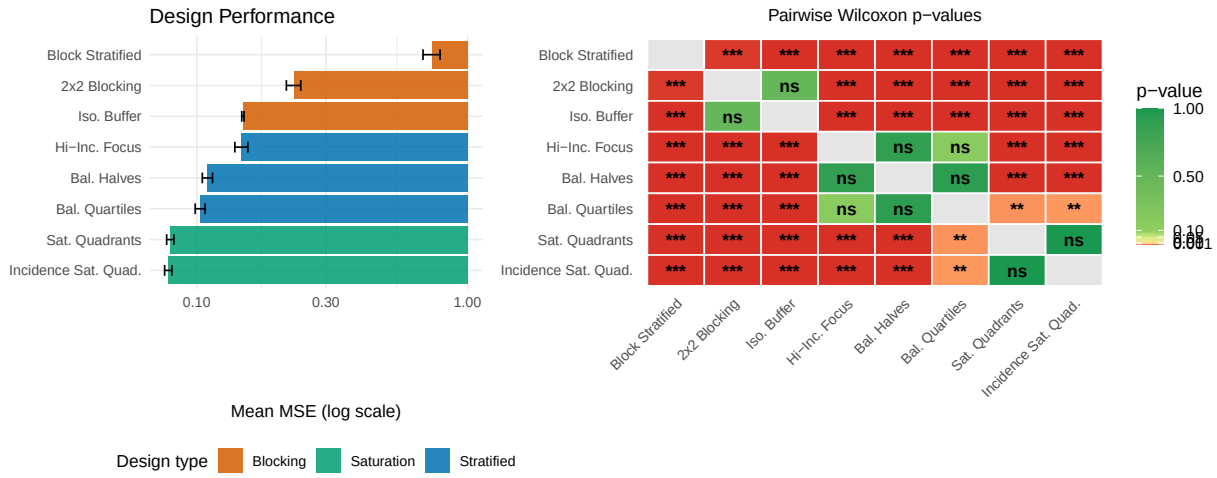

Figure 5: Supplementary comparison of designs at tau = 1.0. Left: mean MSE per design (log scale) with error bars showing ± 1 standard error across scenarios; designs are colored by conceptual group (Blocking, Stratified, Saturation). Right: pairwise Nemenyi p-value heatmap — dark cells indicate statistically significant differences ($p < 0.05$) and light cells indicate non-significant differences. Together these panels show both the magnitude of MSE differences and their statistical significance.

4.3 Sensitivity to Spatial Conditions

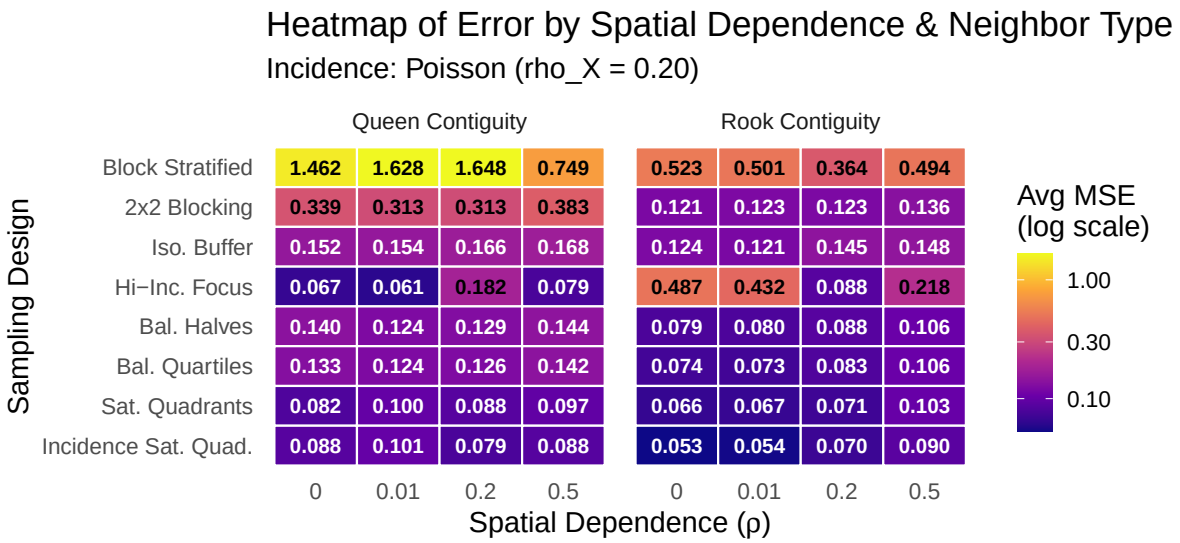

Figure 6: Mean MSE by design and outcome spatial autocorrelation (rho), Poisson incidence (rho_X = 0.20, tau = 1.0). Color uses a log scale to distinguish designs in the low-MSE range; higher rho increases absolute MSE across all designs, but rankings remain stable — Saturation Quadrants and Incidence-Guided Saturation Quadrants consistently achieve the lowest MSE at every rho level.

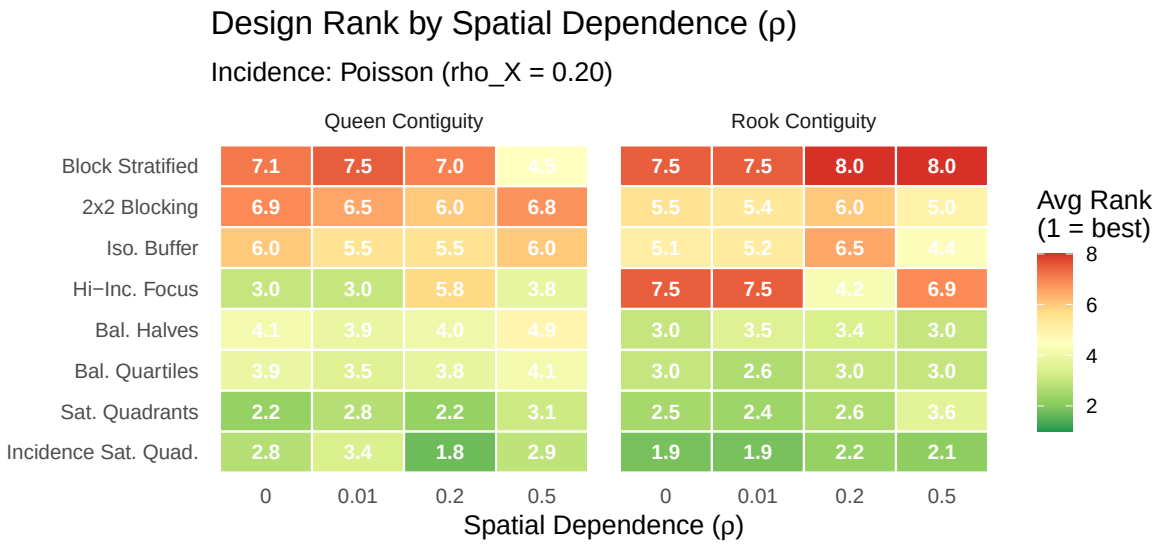

Figure 7: Design rank by outcome spatial autocorrelation (rho), Poisson incidence (rho_X = 0.20, tau = 1.0). Each cell shows the average rank of that design across gamma and spillover direction scenarios at the given rho level (rank 1 = lowest MSE = best). Green indicates top-ranked designs; red indicates the worst. Saturation Quadrants and Incidence-Guided Saturation Quadrants consistently rank 1st and 2nd; Block Stratified Sampling consistently ranks last.

Outcome spatial autocorrelation (ρ) and **spillover magnitude (γ)** both increase absolute MSE — stronger spatial dependence and larger spillover make estimation more difficult for all designs (Figure 6, Figure 7). However, neither parameter changes the design ranking. Notably, the advantage of the saturation designs over mid-tier designs **grows as γ increases**: at $\gamma = 0.8$, the saturation designs' MSE advantage over the stratified and blocking designs is approximately 40% larger than at $\gamma = 0.5$. Saturation-based allocation provides increasing relative returns under stronger spillover — precisely the conditions most likely to occur in real-world spatial interventions.

Two parameters showed **no meaningful impact on design rankings** and are not discussed further:

- **Spillover direction** (`control_only` vs. `both`): both strata yield the same optimal designs (Saturation Quadrants, Incidence-Guided Saturation Quadrants), with modest shifts in absolute MSE magnitude (all conditional Friedman $p < 2.2 \times 10^{-16}$ in both strata).
- **Neighbor type** (rook vs. queen): rankings are invariant; queen contiguity yields slightly lower absolute MSE owing to stronger spatial smoothing with 8 neighbors vs. 4.

4.4 Effect Size Sensitivity and Statistical Power

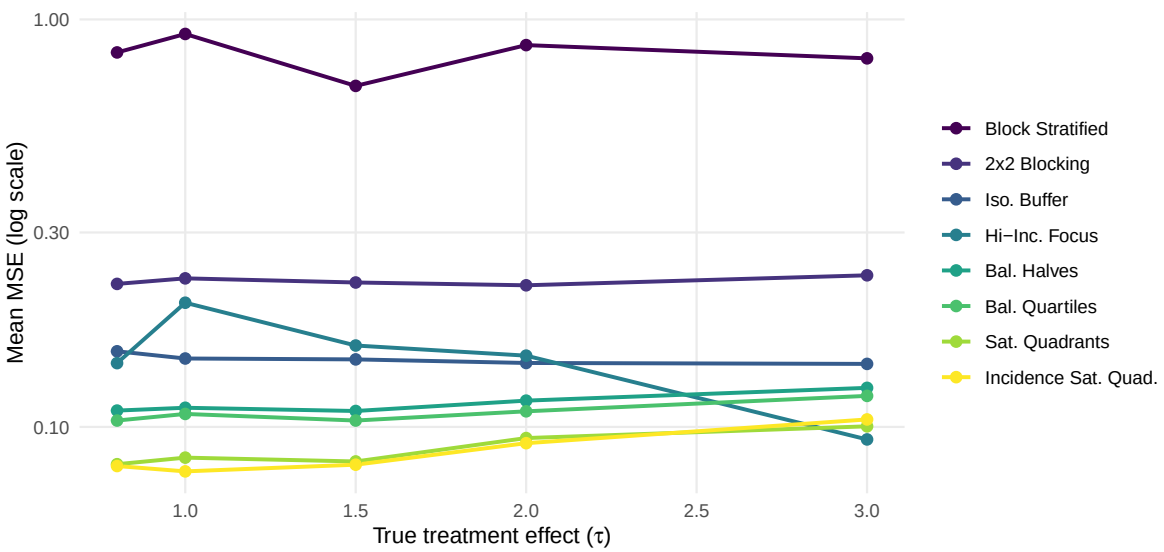

Figure 8: Mean MSE by design across effect sizes (tau in {0.8, 1.0, 1.5, 2.0, 3.0}), Poisson incidence (rho_X = 0.20), log scale. Block Stratified Sampling diverges dramatically at small-to-moderate tau; Saturation Quadrants and Incidence-Guided Saturation Quadrants maintain low MSE across the full range. At tau = 3.0, the large true signal compresses differences among designs.

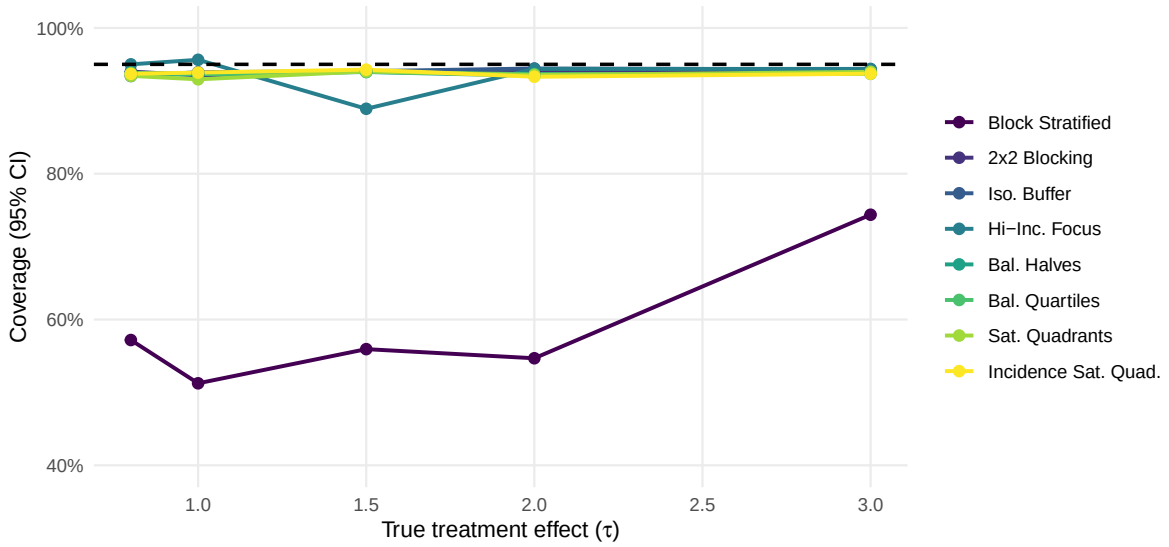

Figure 9: Coverage of 95% confidence intervals by design across effect sizes, Poisson incidence (rho_X = 0.20). The dashed line marks nominal 95% coverage. All designs except Block Stratified Sampling maintain near-nominal coverage throughout; Block Stratified Sampling is severely anti-conservative at all effect sizes (~55% coverage).

Design rankings are **stable across the full effect size range** (Figure 8). At $\tau = 3.0$, a very large treatment signal compresses MSE differences among designs — all designs eventually become adequate when the effect is large enough to overwhelm spatial confounding. At the more practically relevant range $\tau \in \{0.8, 1.0, 1.5\}$, the saturation design advantage is most pronounced and most consequential for trial planning.

Coverage (Figure 9) tells a particularly striking story. All designs except Block Stratified Sampling maintain approximately 94–95% coverage under MLE across all tested effect sizes. **Block Stratified Sampling never exceeds approximately 60% coverage**, regardless of τ. A trial using the Block Stratified Sampling design would produce nominal 95% confidence intervals that contain the true treatment effect less than 60% of the time. This is not simply a power deficit — it is a fundamental structural failure arising from the collinearity between **Z** and **WZ** under the alternating assignment pattern.

5 Design Recommendations and Next Steps

5.1 Recommended Designs

Incidence-Guided Saturation Quadrants is the primary recommendation when reliable baseline incidence data are available. Its quadrant-level saturation rates are set proportional to the observed incidence rank of each region, directly leveraging the spatial structure of the outcome to guide allocation.

Saturation Quadrants is the recommended alternative when baseline incidence estimates are uncertain or unavailable — it randomizes saturation rates across quadrants without requiring in-

cidence data and achieves nearly identical performance (MSE 0.080 vs. 0.079 at $\tau = 1.0$, well within Monte Carlo standard error). Saturation Quadrants is simpler to implement and explain to stakeholders while sacrificing essentially nothing in estimation performance.

Block Stratified Sampling should never be used when spatial spillover is anticipated. Its intuitive appeal — spatially separating treated and control units — is precisely what makes it fail in the presence of spillover. Both saturation designs should be preferred over the blocking and stratified designs (2×2 Blocking, Isolation Buffer, High Incidence Focus, Balanced Halves, Balanced Quartiles) in the SUD application context, where spillover across county lines is plausible.

5.2 Open Questions and Path to Real-World Application

These simulation results establish the theoretical case for saturation-based designs under a regular lattice with uniform cluster populations. The next phase of this work will apply these findings to **non-regular NC county census tract geometries**, raising several methodological questions for the upcoming collaborator meeting:

1. **Spatial weight matrix for irregular polygons:** How should adjacency be defined when clusters are irregular county tracts? Queen contiguity (shared border of any nonzero length) is the natural extension; alternative definitions (distance-based, shared border length-weighted) warrant consideration.
2. **Generalizing the quadrant partition:** Saturation Quadrants and Incidence-Guided Saturation Quadrants rely on rectangular geographic quadrants. For a non-rectangular study region like NC, analogous spatial partitions — k-means geographic clusters, administrative regions, or compass quadrant approximations — would need to replace the regular grid structure.
3. **Relaxing the oracle MLE assumption:** The current estimator uses the true spillover term ($\mathbf{WZ}$) as a covariate. In practice, the spillover structure must be assumed or estimated. Non-oracle MLE — where the spillover term is either omitted or approximated — is a high-priority simulation extension before applying these methods to real data.
4. **Heterogeneous cluster populations:** NC counties vary substantially in population size, which affects expected Poisson event counts per cluster. The current study assumes equal cluster populations ($n = 1{,}000$); variable population sizes may interact with incidence-guided design strategies in non-trivial ways.